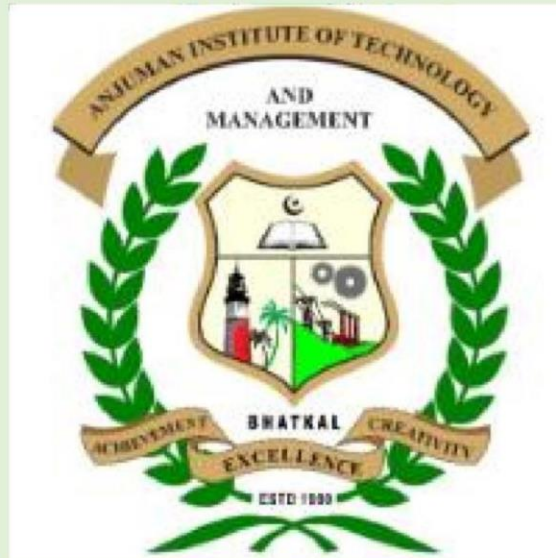


ANJUMAN INSTITUTE OF TECHNOLOGY & MANAGEMENT

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ENGINEERING PHYSICS LABORATORY MANUAL

Name:

Section:

Batch:

USN:

INDEX

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Experiment no :

Date:

OPTICAL FIBER LEARNING OBJECTIVES:

To study light gathering capacity of the given optical fiber in the context of total internal reflection.

AIM:

To determine acceptance angle and Numerical aperture of an optical fiber.

APPARATUS USED:

Single stand plastic optical fibers of different core diameter/length, Laser Source, screen

THEORY:

Numerical aperture represents the light gathering capacity of an optical fiber.

It is given by $N.A = n_o \sin \theta_A = \sqrt{n_1^2 - n_2^2}$

Here n_o is the angle of acceptance

n_1 is the refractive index of the core n_2
is the refractive index of cladding

FORMULA / CALCULATIONS:

$$\theta_A = \tan^{-1}\left(\frac{D}{2L}\right)$$

Where,

θ_A = Acceptance angle ($^\circ$)

D = Diameter of the spot(m)

L = Distance between the screen & the optical fiber end(m)

Numerical aperture is given by $N.A = n_o \sin \theta_A$

Where, n_o = Refractive index of the surrounding medium

PROCEDURE:

1. Connect the fiber to the Laser source.
2. Take the other end of the fiber and project the light output on to the screen to obtain a bright circular spot of size say 5mm.
3. Determine the diameter D of the bright spot and the distance L from the fiber end to the screen and measure the diameter of the spot (D) and the distance between the screen & the optical fiber end (L).
4. Calculate the acceptance angle using the formula

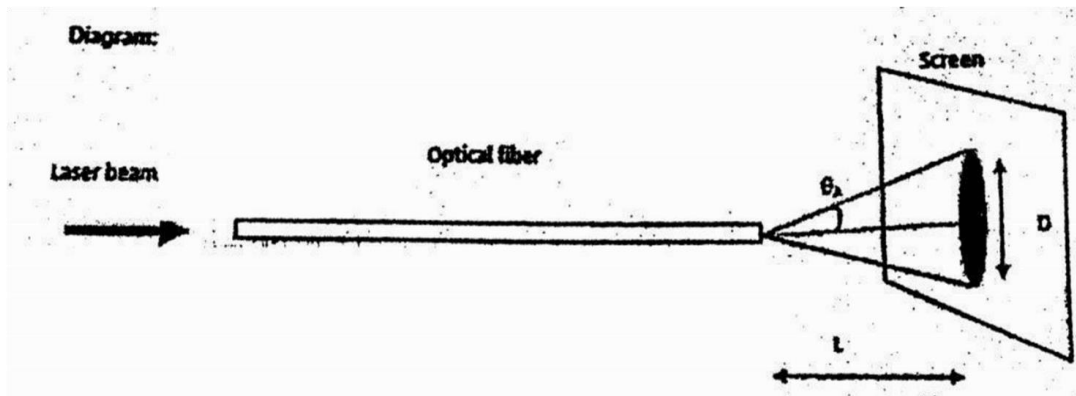
$$\theta_A = \tan^{-1}\left(\frac{D}{2L}\right)$$

5. Numerical aperture is given by $N.A = n_o \sin.\theta_A$

Here n_o is the refractive index of the medium from which light is entering ($n_o=1$ for air)

6. Now for the same fiber repeat this procedure for at least four values of distance L and calculate the acceptance angle and numerical aperture in each case. Finally take the average of the four numerical aperture values.

BLOCK DIAGRAM:



OBSERVATION TABLE:

Distance of laser spot on screen D in mm	Distance between fiber chuck and the screen L (cm) L(mm)		Tan θ_A = $D/2L$	Angle of Acceptance $\theta_A = \tan^{-1}\left(\frac{D}{2L}\right)$	N.A = $n_o \sin \theta_A$ $n_o=1$ for air

Mean θ_A

Mean N.A.

CALCULATIONS:

RESULTS:

1. The acceptance angle of the optical fiber is =
2. The Numerical aperture for the given optical fiber is found to be =

Experiment no :

Date:

WAVELENGTH OF LASER USING GRATING

AIM:

To determine the wavelength of the laser light using diffraction grating

APPARATUS USED:

Laser source, diffraction gratings, grating holder, measuring scale and scale holder.

THEORY:

A transmission diffraction grating is a piece of transparent material on which has been ruled a large number of equally spaced parallel lines. The distance between the lines is called the grating constant, or d .

Light rays that strike the transparent portion of the grating between the ruled lines will pass through the grating at all angles with respect to their original path. If the deviated rays from adjacent rulings, on the grating are in phase, an image of the source will be formed. This will be true when these adjacent rays differ in path length by an integral number of wavelengths of the light. Thus, for a given wavelength λ there will exist a series of angles at which an image is formed. The first time an image is formed will occur when the path difference between adjacent rays from adjacent ruled lines is exactly equal to one wavelength λ . According to Figure that condition will be true at an angle θ such that the equation $d \sin \theta = \lambda$ is satisfied.

The image formed along the original path of the light rays is called zeroth order. For the first order (i.e. $m=1$), there are two images formed symmetrically at different sides of the zeroth order image (see Figure 2).

If the linear distance between the center of the zeroth order image and the first order image is X_m , then the angle θ_m at which the first order image is formed with respect to the original path of the light rays is given by

$$\theta_m = \tan^{-1} (X_m/R)$$

where R is the distance from the grating to the screen. If the grating constant is known, the wavelength of the light is then given by $\lambda = d \sin \theta_m / m$ -----(2)

FORMULA USED: λ

$$\lambda = d \sin \theta_m / m$$

λ is the wavelength of the laser used in meters

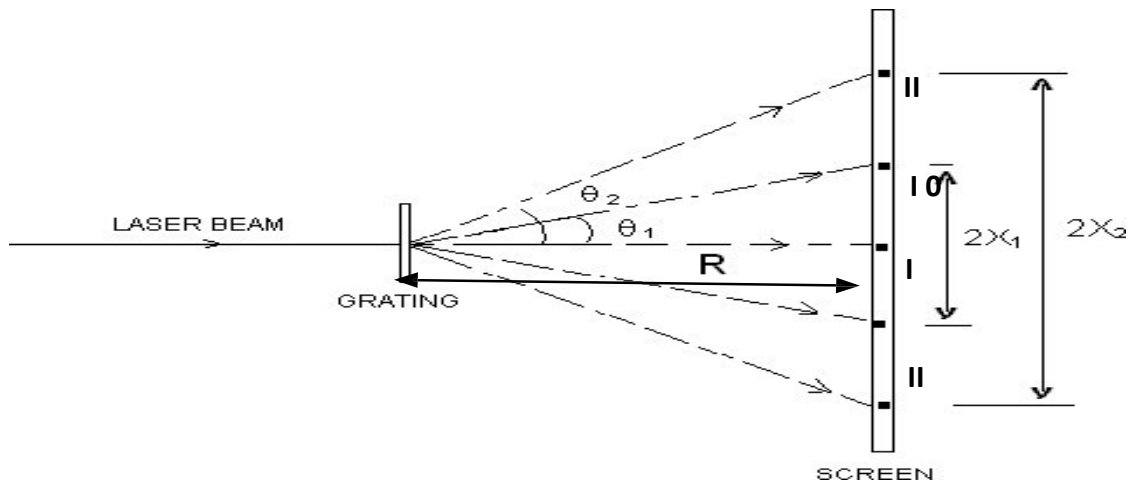
d is the grating constant in meters $\theta_m = \tan^{-1} (X_m/R)$

R is the distance between the grating and screen in meters.

X_m is the distance between the zeroth and first order spots in meters stance in the circuit (m).

PROCEDURE:

1. Place the grating in the holder and the screen is placed at a distance of R as mentioned in the tabular column.
2. The grating is kept between the laser source and the screen.
3. Laser beam undergoes diffraction after passing through the grating. The diffraction spots are observed on the screen.
4. The distances $2X_m$ the symmetrical spots on either side and central bright spots are measured and recorded.
5. The angle of diffraction θ_m is calculated using $\theta_m = \tan^{-1} (X_m/R)$ 6. λ is calculated using the formula $\lambda = C \sin \theta_m / m$

FIGURE:

OBSERVATIONS:

No of lines per inches $n =$ _____ (given)

Grating constant $C = \frac{1}{n} =$ _____ m.

Tabular column:

Trial No	R (in Cm)	Order of the diffraction pattern (m)	$2X_m$ (in Cm)	X_m in (cm)	Diffracting angle $\theta_m = \tan^{-1} (X_m/R)$	$\lambda = C \sin \theta_m / m$

Mean $\lambda =$

CALCULATIONS:

By Calculation

Grating constant $C =$ _____ m

Mean value of $\theta_m =$ _____ degrees

Wavelength of laser $\lambda = C \sin \theta =$ _____ m.

RESULT:

The Wavelength of the given laser source isnm



Experiment no :

Date:

**DETERMINATION OF DIELECTRIC CONSTANT BY CHARGING AND DISCHARGING
METHOD**

AIM: To determine the dielectric constant of the given dielectric material by the method of charging and discharging.

APPARATUS: Capacitor, Resistor, Two way toggle switch, Voltmeter, stop watch.

THEORY:

Capacitors are devices which store electric energy by means of an electrostatic field and release this energy later. The voltage across the capacitor when it gets charged gradually at any instant of time t , is given by

$$V_c = V (1 - e^{-t/RC}),$$

where V is the voltage applied, R is the resistance and C the capacitance in the circuit. While discharging through the resistor R , the capacitor voltage at any instant t is given by,

$$V_c = V e^{-t/RC}$$

Let $T_{1/2}$ be the time required to charge or discharge a capacitor to 50%.

$$\text{When } t = T_{1/2}, V_c = V/2$$

$$V/2 = V e^{-t/RC} \quad t/RC =$$

$$2 \quad t/RC = \log_e 2 = 0.693$$

$$C = T_{1/2} / 0.693 R$$

But $C = \epsilon_0 A / d$ where A and d are the thickness and area of the dielectric material. ϵ_0 is the permittivity of free space and K the dielectric constant.

$$K = \frac{1.44 \times 10^{-6} \times T_{1/2} \times d}{\epsilon_0 A R}$$

FORMULA: Dielectric constant of a given material is given by

$$K = \frac{1.44 \times 10^{-6} \times T_{1/2} \times d}{\epsilon_0 A R}$$

Where, K = Dielectric constant of the material

$T_{1/2}$ = Time taken by the capacitor for half charging / discharging in S.

d = Distance between the two plates = m.

ϵ_0 = Permittivity of free space = 8.852×10^{-12} F/m

A = Area of the capacitor plate = m^2 .

R = Resistance = Ω .

[Constants for Kit-1: $A = 47 \times 5 \times 10^{-6} m^2$, $C = 0.01 \mu F$, $d = 0.075 \times 10^{-3} m$, $R = 100 \times 10^3 \Omega$]

PROCEDURE:

1. The circuit connections are to be made as shown in the figure. Select the resistance R(100K Ω) and capacitor C1.
 2. The digital stop clock is to be reset by pressing the reset button. The display indicates 00.00 seconds.
 3. Switch S1(charge-discharge) has to be turned to the charge position and switch S2 (haltstart) has to be turned to the start position. Start watching the digital stop clock and voltmeter simultaneously.
 4. Immediately the values of voltage are noted at an interval of 5 sec without pause in between, until the capacitor is fully charged.
 5. When the capacitor is charged to maximum voltage (4.5 and above), the charging is stopped and the charge discharge switch is turned to the discharge position. The clock is reset.
 6. The voltage across the discharging capacitor is to be noted at an interval of 5 seconds. This is done until the capacitor is discharged fully.
 7. Draw a graph by taking time on x axis and voltage on y axis as shown in the graph. The charging and discharging curve intersects at a point, where the voltage across the capacitor during charging and discharging remains the same. This time has to be noted
 8. Dielectric constant is determined using the formula
-

CIRCUIT DIAGRAM:

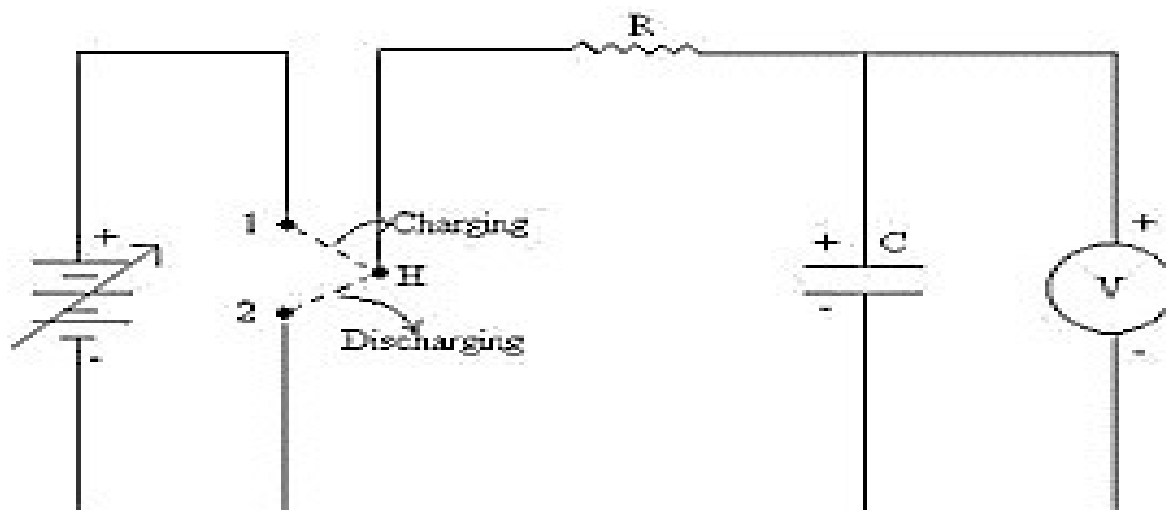
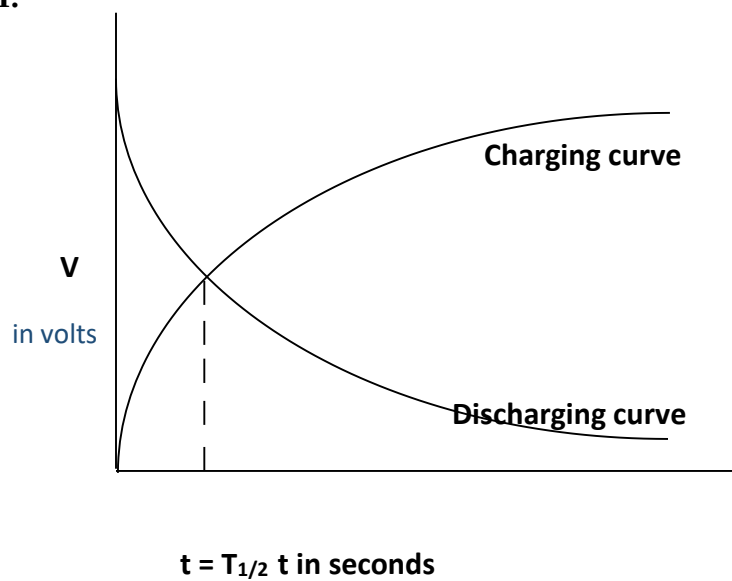


Figure 1

GRAPH:



TABULAR COLUMN:

Time 't' in second	Voltage during charging 'V' in volts	Voltage during discharging 'V' in volts

CALCULATIONS:

The Value of Dielectric Constant K=

RESULT: Dielectric constant (K) of the given material is found to be -----

Experiment no :

Date:

PHOTODIODE CHARACTERISTICS AIM:

- 1) To study the reverse bias characteristics in illuminated condition.
- 2) To study the variation of photocurrent with the intensity of illumination.

APPARATUS USED:

Photodiode fixed in a box, regulated power supply, current meter, voltmeter, light source

THEORY:

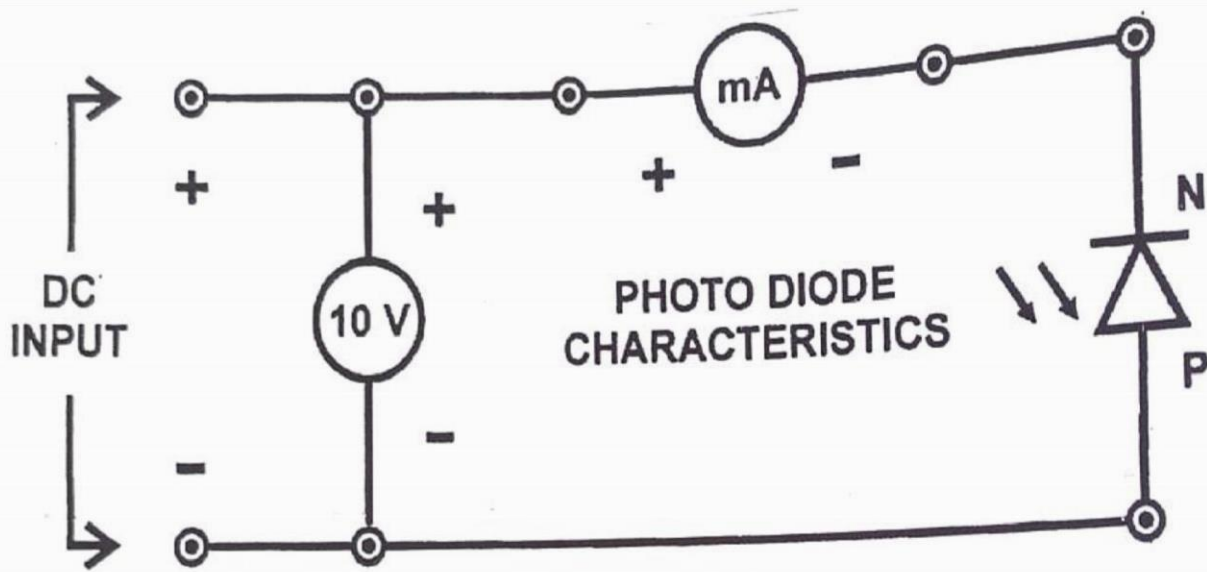
The construction of the Photodiode light sensor is similar to that of a conventional PNjunction diode except that the diodes outer casing is transparent so that light can fall upon the junction. All PN-junctions are light sensitive and can be used in a photoconductive (PC) mode. The PN-junctions of the photodiode is always reverse biased so that only the diode leakage current or dark current can flow. This reverse bias condition causes an increase of the depletion region which is the sensitive part of the junction.

When light falls on the junction more hole/electron pairs is formed and the leakage current increases. The leakage current increases as the illumination of the junction increases. Diode current is directly proportional to light intensity.

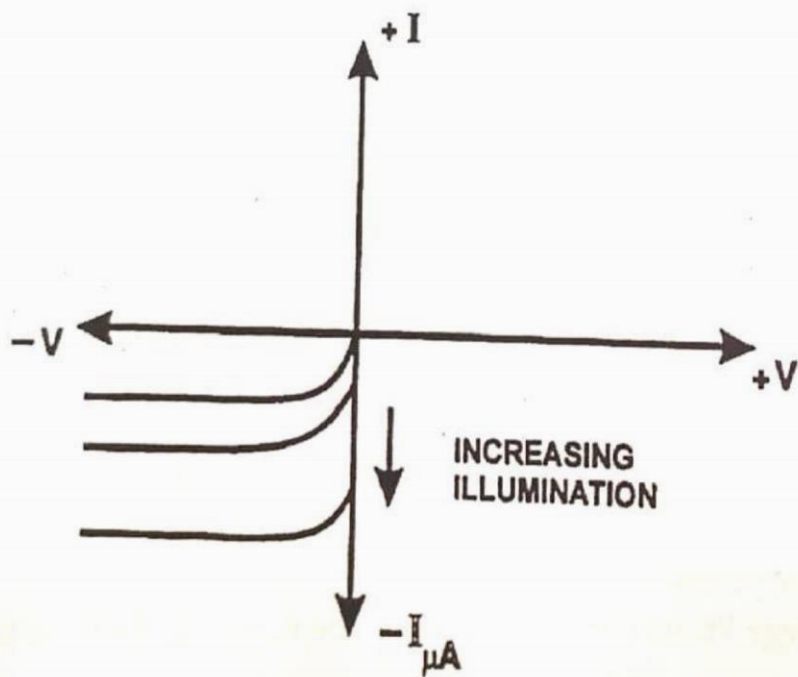
PROCEDURE:

1. The regulated power supply is connected to photo diode in reverse bias condition. Current meter is connected in series. The voltmeter is connected across the photodiode.
 2. The illuminated source of light is kept at a particular distance (about 20 cm.) from the photodiode.
 3. The reverse bias voltage is varied from zero to 5 volts in steps of 0.1 up to 1 volt and in-step of 1 volt up to 5 volt. the corresponding current meter readings are noted.
 4. To study the variation of photocurrent with the intensity of illumination the reverse bias voltage is set to 5volts. The photodiode is kept at different distances from the source of light and the corresponding current is noted in each case.
 5. Plot the graph of reverse voltage Vs reverse current across the device as shown in the graph.
-
-

FIGURE:



(Fig-1)



TABULATIONS:

Illumination (I_1) =cm		Illumination (I_2) =cm	
V_R (V)	I_R (mA)	V_R (V)	I_R (mA)

RESULT:

The VI characteristics of a photodiode and the variation of the photo current as a function of reverse voltage and intensity is studied.

Experiment no :**Date:****FERMI ENERGY**

AIM: To find the Fermi energy of the copper by studying the resistance variation at different temperatures.

APPARATUS REQUIRED: DC regulated power supply, milli voltmeter, heating arrangement, thermometer, copper wire etc.

THEORY:

Fermi energy is given by $E_F = \frac{1}{2} m v_F^2$ where v_F is the Fermi

velocity, m is the mass of the electron.

$$\sigma = n e^2 \lambda_F / m v_F$$

Where n is the electron concentration, e is the charge on the electron, λ_F is the mean free path $v_F = n e^2 \lambda_F / m \sigma$ $\sigma = L/RA$ where L is the length, A is the area of the specimen wire and R is the resistance.

$$\begin{aligned} \text{Therefore } v_F &= n e^2 \lambda_F R A / m L \\ &= n e^2 \lambda_F T R A / m L T \end{aligned}$$

$$\begin{aligned} \text{Therefore } E_F &= \frac{1}{2} m v_F^2 \\ &= \frac{1}{2} m (n e^2 \lambda_F T R A / m L T)^2 \\ &= \frac{1}{2} m (n e^2 X R \pi r^2 / m L T)^2 \text{ where } X = \lambda_F T \text{ (constant) and } A = \pi r^2 \text{ where } r \text{ is the} \\ &\text{radius of the wire.} \end{aligned}$$

$$E_F = \left[\frac{n e^2 \pi A r^2}{L \sqrt{(2m)}} \right]^2 \left[\frac{\Delta R}{\Delta T} \right]^2 \dots \dots \dots \text{Joules}$$

FORMULA USED:

$$E_F = \left[\frac{n e^2 \pi A r^2}{L \sqrt{(2m)}} \right]^2 \left[\frac{\Delta R}{\Delta T} \right]^2 \times \frac{1}{1.6 \times 10^{-19}} \dots \dots \dots \text{eV}$$

.

$$\left[\frac{\Delta R}{\Delta T} \right]$$

is the slope of the graph

m is the mass of electron (m) e is the charge of electron (e) r is the radius of the wire (r)

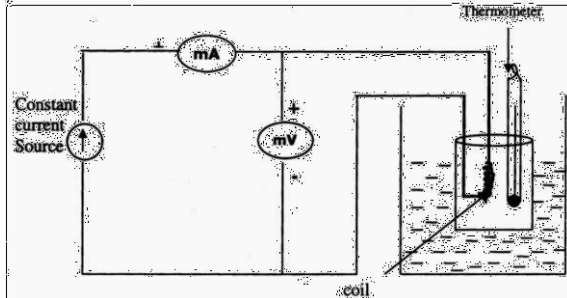
n is the number density of electrons (n)

L is the length of the wire (L)

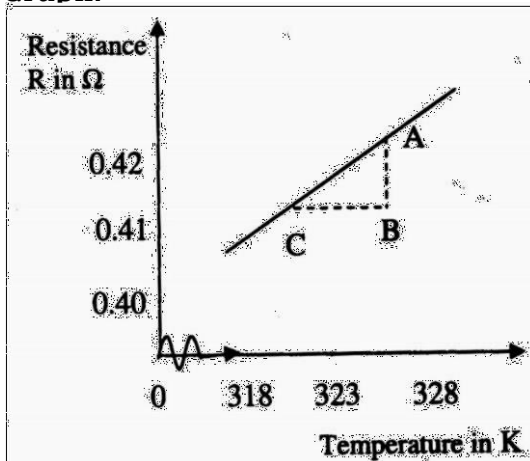
PROCEDURE:

1. The length and radius of the wire is noted.
 2. Circuit connections are made as shown in the figure.
 3. The wire is wound over an insulating tube to form the coil.
 4. The coil is immersed in water.
 5. The power supply is switched on and voltage and current are noted for the lab temperature.
 6. The water is heated till water boils.
 7. Voltage and current are noted for different temperatures while cooling.
 8. The graph is drawn taking temperature in degree Kelvin along X-axis and resistance along Y axis as shown in the figure.
 9. The slope of the straight line is calculated.
 10. Fermi energy of the copper is calculated using the formula.
-
-

FIGURE:



Graph:

**OBSERVATION:**

m is the mass of electron (m)kg

e is the charge of electron (e)C r is the

radius of the wire (r)m n is the

number density of electrons (n).....

L is the length of the wire (L).....m

TABULAR COLUMN :

Temperature (°C)	Temperature (K)	Voltage (mV)	Current (mA)	Resistance (Ω)

CALCULATIONS:

Slope =

$$E_F = \left[\frac{ne^2 \pi A r^2}{L \sqrt{2m}} \right]^2 \left[\frac{\Delta R}{\Delta T} \right]^2 \times \frac{1}{1.6 \times 10^{-19}} \dots\dots\dots \text{eV}$$

$$= \dots\dots\dots \text{eV}$$

RESULT: Fermi energy the given material = $\dots\dots\dots$ eV



Experiment No:

Date:

ENERGY BAND GAP OF A SEMICONDUCTOR BY FOUR PROBE METHOD

Aim: To determine the energy gap of a semiconductor by using four probe method.

Apparatus: Semiconductor in the form of a crystal, thermometer, four probes apparatus

THEORY:

Many conventional methods of measuring resistivity are unsatisfactory for semiconductors because metal semiconductor contacts are usually rectifying in nature. Also there is minority carrier injection by one of the current carrying contacts. An excess concentration of minority carriers will affect potential of the other contacts. An excess concentration of minority carriers will effect potential of the other contacts and modulate the resistance of the material. The method described here overcomes the difficulties mentioned above and also offers several other advantages. It permits the measurement of the resistivity of the sample having the wide variety of shapes, including the resistivity of small volume within bigger pieces of semiconductor. In this way the resistivity of both the PN junction can be determined with good accuracy.

FORMULA: The resistivity of the material of the crystal is given by

$$\rho = \frac{V}{I} \times 2\pi S$$

where, S is a distance between two probes

$$Resistance R = V/I$$

R is the resistance of the the crystalline ohm,

V is the voltage in mV

I is the Current in mA

Energy gap of the semiconductor is
given by

$$E_g = \frac{2.303 \times 2 \times K \times \log_{10} R}{T-1}$$



$$= 2.303 \times 2 \times K \times \text{Slope}$$

Where,

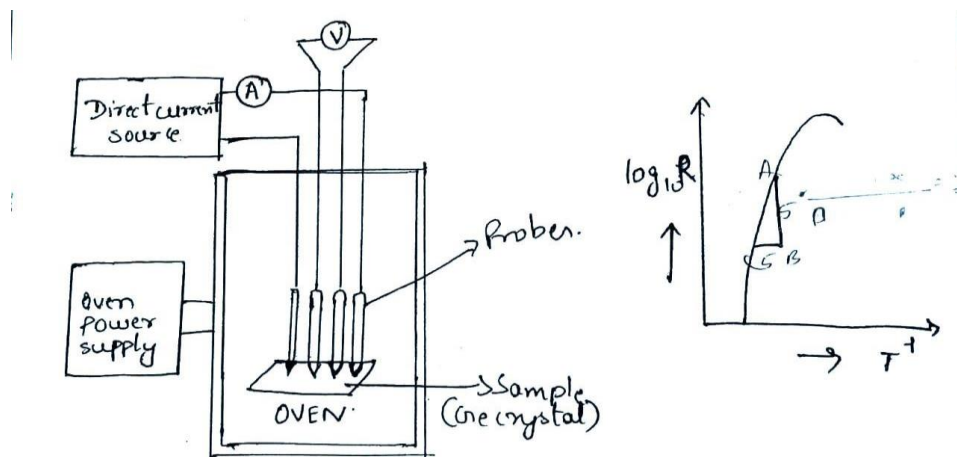
k is Boltzmann constant = 8.625 eV/K

Slope is calculated from the graph of $\log_{10}$ vs T^{-1}

PROCEDURE:

1. Connect the circuit as in diagram.
2. Immerse the thermistor in the port in the crystal holder of the four probes apparatus.
3. Adjust the current at 5.00 mA. This value of current should be kept constant.
4. Switch on the oven and heat the sample up to 200°C .
5. Note down the value of voltage for every 10°C increase in temperature starting from 200°C .
6. Calculate the resistance and resistivity of a semiconductor material.
7. Plot the graph of $\log_{10}$ versus $(1/T)$ and calculate its slope.
8. Calculate the energy gap of the semiconductor using the given formula.

FIGURE:



OBSERVATIONS:

Distance between two probes (S) = $\frac{0.2}{5.89} = 0.0339\text{cm}$ (Here 5.89 is correction factor)

Current, I = 5.00 mA

TABULAR COLUMN:

Sl. No.	Temp T, K	Voltage in mV	Resistance $R = V/I \Omega$	Resistivity $\rho = \frac{V}{I} \times 2\pi S \Omega\text{-m}$	$[(1/T) \times 10^{-3}]$ K-1	$\log_{10} \rho$

CALCULATIONS:

$E_g = 2.303 \times 2 \times K \times \text{Slope}$

Slope is calculated from the graph of $\log_{10} \rho$ vs T^{-1}

RESULT:

The energy gap of the given semiconducting material is $E_g = \underline{\hspace{2cm}}$ eV

Experiment No:**Date:****PLANCK'S CONSTANT****AIM:**

To determine the planck's constant by using LED

APPARATUS USED:

Power supply, miliammeter, DC volt meter and LED

THEORY:

When an electron of charge 'e' is accelerated through a potential difference of V volts then the electron acquires energy equal to eV. When an LED is forward biased the electrons from then-region recombine with the holes in the p-region and emit light of frequency $f = c / \lambda$. Hence the energy of an electron accelerated across the potential barrier of the LED is $eV = h\nu$ where h is the Planck's constant. By finding the knee voltage and the wavelength of light emitted, Planck's constant can be calculated.

FORMULA:

Planck's constant: $h = eV \lambda / C$ Js

Graphical method: $h = e (\text{slope})$ Js

Where, e -charge of electron = $1.6 \times 10^{-19} \text{C}$

V - knee voltage of the given LED λ -

Wavelength of light emitted by LED ($\text{\AA}$)

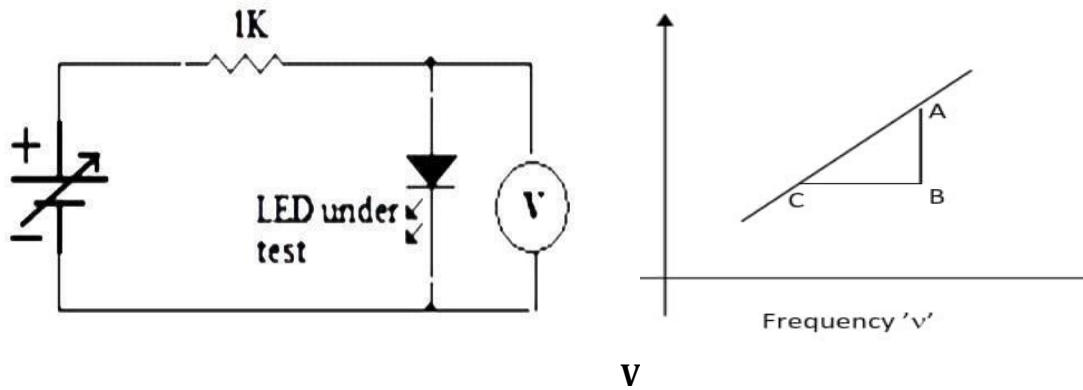
C -velocity of light = $3 \times 10^8 \text{ms}^{-1}$

PROCEDURE:

The circuit connections are made as shown in the diagram. An LED of given wavelength. is forward biased and the corresponding knee voltage is noted. The experiment is repeated for LED's of different wavelengths and the corresponding knee voltage is noted in each case. The frequency of the light emitted by LEDs' of different wavelengths is calculated and tabulated. The Planck's constant is calculated from the tabulated values of knee voltage and wavelength of each LED and hence the average. value of the Planck's constant is calculated. A graph is plotted between frequency on x-axis and knee Voltage on y-axis. The points are connected to a Straight line and the

slope of the straight line (m) is determined. The Planck's constant is determined by using the formula $h = e \times \text{slope}$.

FIGURE :



TABULAR COLUMN:

Colour	WaveLenght λ	frequency $f = c / \lambda$ (Hz)	Knee Voltage (V)	$h = e \nu \lambda / C$

CALCULATIONS:

Planck's constant: $h = eV\lambda / C$ Js

Graphical method: $h = e$ (slope)Js

RESULT: The Planck's constant , h

Experimental method =Js

Graphical method =Js



Experiment No:**Date:****ENERGY GAP OF A SEMICONDUCTOR****AIM:** To determine the forbidden energy gap of a semiconductor.**APPARATUS:** P-N diode, DC regulated power supply, Voltmeter, Milliammeter, 1 KW resistor, Beaker, Thermometer, Heater.**PRINCIPLE:** The forbidden energy gap of a material is a collection of energy levels above the top of the valence band and below the bottom of the conduction band. For the determination of the energy gap of a semiconductor, a semiconductor in the form of P-N diode is used. The forward current in such a diode is given by, $I_f = I_s[\exp(eV/hkBT) - 1]$

Where V = forward voltage across the junction, e = electronic charge, k_B = Boltzmann constant, T = absolute temperature, I_s = reverse saturation constant and h = a constant called emission coefficient I_s is given by, $I_s = BT^3 \exp(-E_g / k_B T)$ where B is a constant.

For a low constant forward current, the above equations can be approximated to yield an equation, $eV = E_g - h k_B T$

Hence a plot of V versus T gives a straight line with the V -intercept equal to E_g/e at $T = 0K$. From this, the energy gap of the semiconductor can be obtained.

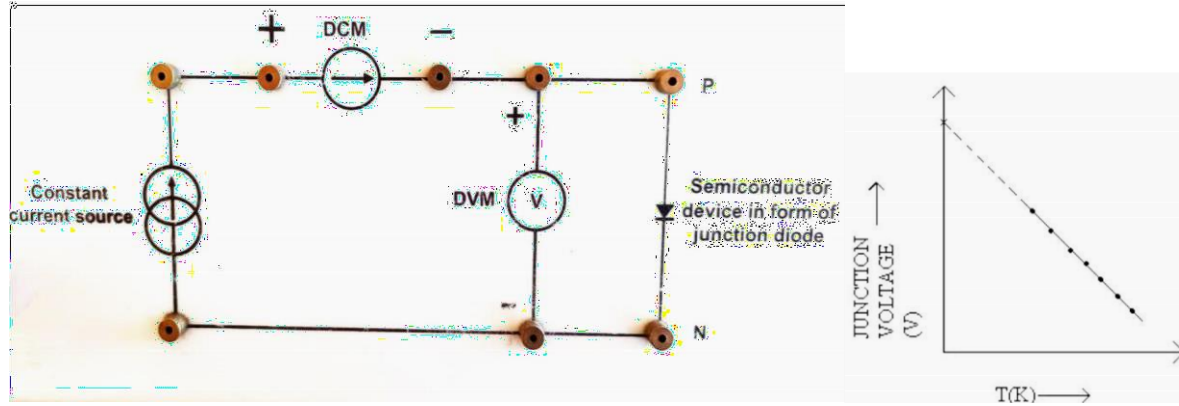
FORMULA: **$E_g = Y \text{ intercept} = \dots\dots\dots eV$** **$E_g = e \times Y \text{ intercept} = \dots\dots\dots J$** e -charge of electron = $1.6 \times 10^{-19} \text{ C}$ **PROCEDURE:**

The circuit is built up as shown in figure. The forward biased voltage is kept at room temperature. A constant forward current is passed through the diode and the voltage developed across the diode at this temperature is noted. Then the diode is immersed in hot water bath. Voltage across the diode is noted down for different temperatures as water bath cools down. The readings are noted down till the water bath attains room temperature.

A graph with voltage along Y-axis and temperature along X-axis is drawn. The Y-intercept of the graph at $0K$ is found. From this the energy gap of the semiconductor is calculated.



FIGURE :



TABULAR COLUMN:

Temperature		Junction voltage	
$^{\circ}\text{C}$	K	Si	LED

CALCULATIONS:

For Si : $E_g = Y \text{ intercept} = \dots\dots\dots \text{eV}$

$E_g = e \times Y \text{ intercept} = \dots\dots\dots \text{J}$

For LED: $E_g = Y \text{ intercept} = \dots\dots\dots \text{eV}$

$E_g = e \times Y \text{ intercept} = \dots\dots\dots \text{J}$

Result: Energygap of given semiconductor (Si) $E_g = \dots\dots\dots \text{J}$

Energygap of given semiconductor (LED) $E_g = \dots\dots\dots \text{J}$



Experiment No:

Date:

TRANSISTOR CHARACTERISTICS

AIM: To study the input, output and transfer characteristics of an N-P-N transistor in the common emitter mode and also determining the input resistance (R_i) and the current gain factor (β) of the given transistor.

APPARATUS: Given transistor (NPN), variable DC power supplies (0-1V&0-10V) DC micro ammeter (0-200 μ A), DC milli ammeter (0-10 mA), DC voltmeter (0- 1V&0-10V) and connecting wires.

THEORY:

Transistor is the semiconductor with three terminals and two junction device. The three terminals are emitter, base and collector. Emitter is heavily doped and its function is to emit or eject electrons to the base region. The Base is lightly doped and its function is to pass most of the electrons emitted by the emitter to the collector region. The collector is intermediately doped and it collects the emitter ejected electrons and hence has larger area. Since transistor has three terminals one terminal is made common to both input and output. According there are three possible configuration. If emitter is made common then the configuring is called common emitter configuration. Common emitter configuration is most efficient because of its high current gain, high voltage and power gain. In common emitter configuration current gain is the ratio of change in the collector current to the change in the base current. i.e. $\beta = \Delta I_c / \Delta I_B$. Transistor characteristic curves are the graphs which show the relationship between the various currents and voltages. Input characteristic curve is the curve between the base current I_B and the base-emitter voltage V_{BE} at constant collector - emitter voltage V_{CE} . In CE configuration output characteristic is the curve between the collector current I_c and the collector emitter voltage V_{CE} at constant base current I_B .

FORMULA:

1) Input resistance $R_i = \Delta V_{BE} / \Delta I_B$

Where ΔV_{BE} = Change in the base emitter voltage in volt

ΔI_B = Change in base current

3) Current gain $\beta = \frac{I_{C2} - I_{C1}}{I_{B2} - I_{B1}}$

β is the current gain

I_{C1} and I_{C2} are the maximum collector currents corresponding to base currents I_{B1} and I_{B2} respectively.

3) Current amplification factor $\beta = \Delta I_c / \Delta I_B$

$\Delta I_c / \Delta I_B$ is the slope from the graph.

PROCEDURE:

- The common emitter circuit for studying the transistor characteristics of a NPN transistor is shown in fig. First identify the terminals of different devices required for the experiment on the experimental box.
- Give the connections using connecting wires carefully according to the circuit diagram. Before switching on the circuit, verify once again the circuit connections. Now turn all power supply knobs to the minimum position & switch on the power supply. Check that circuit is working properly.

Input characteristics:

- To study the input characteristics of the transistor first turn all power supply knobs to minimum position. Now collector-Emitter voltage V_{CE} is set to 2 volt by varying V_{BE} . Keeping $V_{CE} = 2$ volt as constant vary the Base-Emitter voltage V_{BE} is increased from zero and apply V_{BE} as 0.1, 0.2, upto 0.9 corresponding base current I_B is noted down in the table provided
- A graph of V_{BE} along X- axis & I_B along Y- axis is plotted. The straight line portion of the curve extended backward to meet with the x axis at V_k where V_k represents the Knee Voltage. Slope of the curve is found in active region of transistor which is (i.e. linear portion of the curve) the reciprocal of the input resistance values.

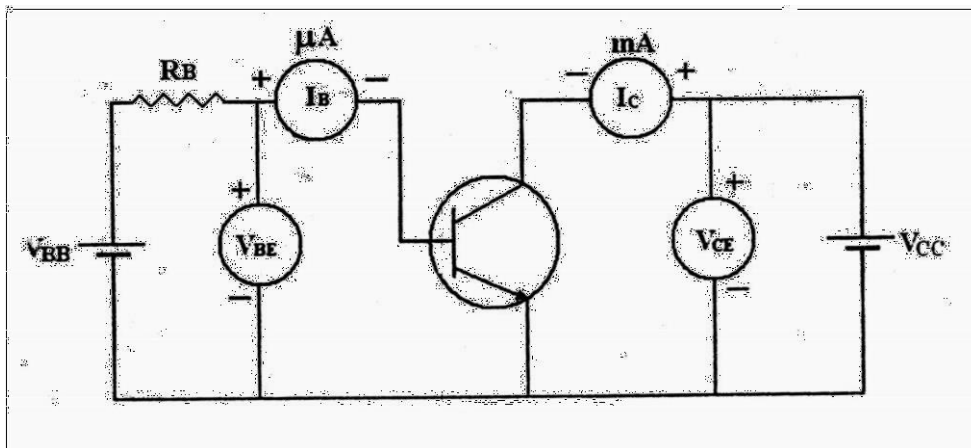
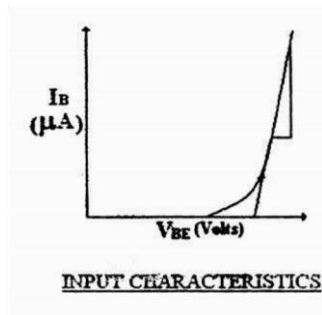
Output characteristics:

- To study the output characteristics of the transistor. Again turn all the power supply knobs to minimum position. Now the Base current I_B is set to $20 \mu A$ by turning V_{BB} knob. Keeping $I_B = 20 \mu A$, apply V_{CE} values as 1, 2, 3, volts till 8 volt and note down corresponding I_c values in milli amperes each time. Tabulate all the values in relevant tabular column for output characteristics. Care should be taken that while taking each reading of I_c , I_B should read the constant values i.e. $I_B = 20 \mu A$.
- Apply I_B values as $20 \mu A$, $60 \mu A$, & $100 \mu A$ & note the I_c values in milli amperes each time. Tabulate the readings in relevant tabular column for transfer characteristics.
- A graph of I_B along X-axis & I_c along Y-axis is plotted. The graph obtained will be a straight line, calculate the current gain $\beta = \Delta I_c / \Delta I_B$.



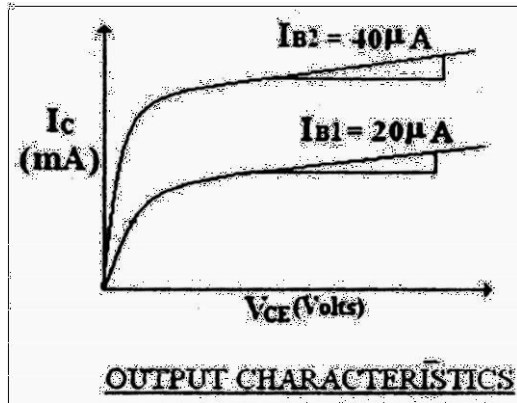
Transfer characteristics:

- To study the Transfer characteristics of the transistor. Turn all the power supply knobs to minimum position again. Now set the collector-Emitter voltage $V_{CE} = 2$ volt.
- Apply I_B values as $10\mu A$, $20\mu A$, $30\mu A$ till $100\mu A$ & note the I_C values in milli amperes each time. Tabulate the readings in relevant tabular column for transfer characteristics.
- A graph of I_B along X-axis & I_C along Y-axis is plotted. The graph obtained will be a straight line, calculate the slope and slope will be the ratio $\Delta I_C / \Delta I_B$, and hence the current gain β values.

CIRCUIT DIAGRAM:**OBSERVATIONS:****Input characteristics:**

$V_{CE} = 2V$	
$V_{BE}(mV)$	$I_B(\mu A)$
0	0
0.1	
0.2	
0.3	
0.4	
0.5	
0.6	
0.7	
0.8	
0.9	

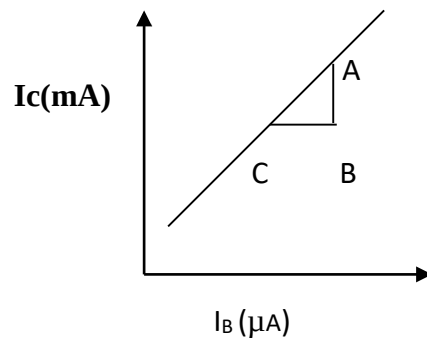
Output characteristics:



$I_B = 20 \mu A$		$I_B = 60 \mu A$		$I_B = 100 \mu A$	
$V_{CE}(V)$	$I_C(mA)$	$V_{CE}(V)$	$I_C(mA)$	$V_{CE}(V)$	$I_C(mA)$
0	0	0	0	0	
1		1		1	
2		2		2	
3		3		3	
4		4		4	
5		5		5	
6		6		6	
7		7		7	
8		8		8	
9		9		9	

Transfer characteristics:

$V_{CE} = 2$ volt	
I_B (μA)	I_C (mA)
10	
20	
30	
40	
50	
60	
70	
80	
90	
100	



CALCULATION:

1) Input Characteristics:

(i) Knee Voltage=_____V

(ii) Input resistance $R_i = \frac{\Delta V_{BE}}{\Delta I_B} =$ _____ Ω

2) Output Characteristics:

(i) DC current gain $\beta = \frac{I_{C2} - I_{C1}}{I_{B2} - I_{B1}} =$

3) Transfer Characteristics:

(i) Current amplification factor $\beta = \Delta I_C / \Delta I_B$ **RESULT:**

(i) Knee Voltage=_____V

(ii) Input resistance $= \frac{\Delta V_{BE}}{\Delta I_B}$ $R_i =$ _____ Ω (iii) DC current gain $\frac{I_{C2} - I_{C1}}{I_{B2} - I_{B1}} = \beta =$ (iv) Current amplification factor $\beta = \Delta I_C / \Delta I_B$ 

Experiment no :

Date:

FIELD ALONG THE CIRCULAR COIL

AIM: To study the variation of magnetic field along the axis of the circular coil.

APPARATUS: Digital Stewart-Gee apparatus, AC power supply.

THEORY: A stationary electric charge produces an electric field where as moving electric charge produces magnetic field around it. This is the basic principle of electromagnetic induction. The magnetic field at the center of the coil carrying current 'I' is given by,

$$B_x = \frac{\mu_0 n I}{2R} \left[\frac{R^2}{(R^2 + X^2)^{3/2}} \right]$$

Where n – Number of turns in the TG coil,

R– Radius of the coil (cm)

x – Distance between the center of the coil and pointer in the compass box

μ_0 - Permeability of free space = $4\pi \times 10^{-7} \text{ Hm}^{-1}$.

I – the current through the coil (I)

When the coil is placed with its vertical plane in the magnetic meridian, the direction of the magnetic flux density B is perpendicular to magnetic meridian. Hence the needle of the deflection magnetometer placed at any point shows a deflection such that

$B_x = B_H \tan \theta$, Where B_H – horizontal component of earth's magnetic field

FORMULA:

$$B_x = \frac{\mu_0 n I}{2R} \left[\frac{R^2}{(R^2 + X^2)^{3/2}} \right]$$

Where n – Number of turns in the TG coil,

R– Radius of the coil (cm)

x – Distance between the center

of the coil and pointer in the compass box μ_0 - Permeability of free space = $4\pi \times 10^{-7} \text{ Hm}^{-1}$.

I – the current through the coil (I)



2

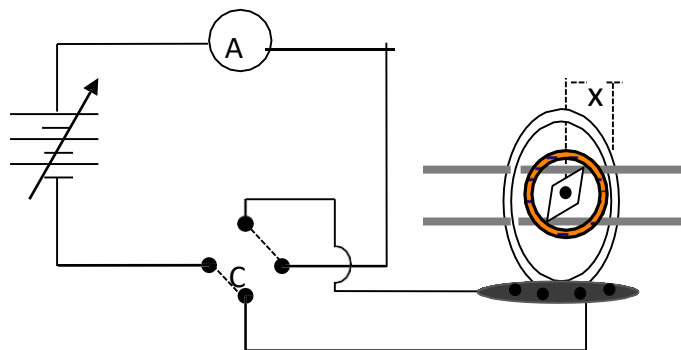
$$B_x = B_H \tan \theta$$

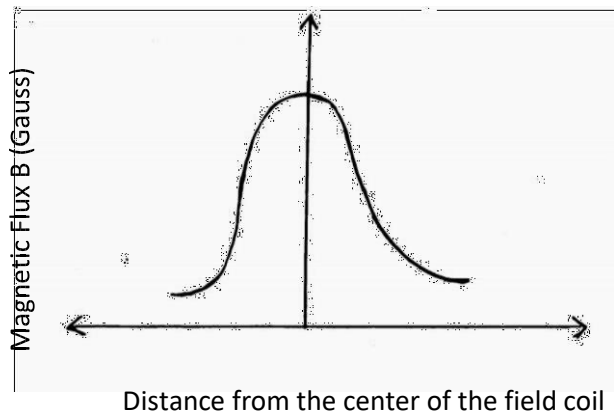
Where B_H – horizontal component of earth's magnetic field
 θ – mean deflection in TG.

PROCEDURE:

1. The connections are made as shown in the circuit diagram.
2. Arrange the deflection of the magnetometer in the magnetic meridian of the earth
3. Now align the plane of the coil with respect to 90° - 90° line of the magnetometer.
4. Turn the whole board such that pointer in the compass reads 0-0
5. Keep the magnetometer exactly at the center of the coil (for this case $x = 0$).
6. Pass a current I (say 50mA) to flow through the coil and the corresponding magnetometer deflections θ_1 and θ_2 are noted.
7. The direction of the current is reversed by using the commutator C and the corresponding magnetometer deflections θ_3 and θ_4 are noted.
8. Average deflection θ is calculated.
9. Calculate the magnetic field at the center of the coil by using the given formula
10. Repeat the experiment for different values of x (say 5cm, 10cm, ...) by sliding the magnetometer along the axis.
11. Find the average of both B and B_H .

DIAGRAM:



NATURE OF THE GRAPH:**OBSERVATIONS:**

No. Of turns of the coil, $n = 140$

Current through the coil, $I = 0.05 \text{ A}$

Radius of the coil, $r = \text{circumference} / 2\pi = 84 / 2\pi = 0.1336 \text{ m}$

μ_0 - Permeability of free space $= 4\pi \times 10^{-7} \text{ Hm}^{-1}$.

Horizontal component of earth's magnetic field $B_H = 3.5 \times 10^{-5} \text{ T}$

Sl. No.	Distance from the center, X in m	Deflections in degrees				Average θ in degree	Tan θ	B _x In T	B _x = B _H Tan θ In T
		θ_1	θ_2	θ_3	θ_4				
1									
2									
3									
1									
2									
3									

Mean value of Experimental B_x = (T)

Mean value of Theoretical B_x = (T)

CALCULATION:

1) The magnetic field intensity at the center of the coil B_x

Experimental B_x =(T)

Theoretical B_x =(T)

RESULT:

1) The magnetic field intensity at the center of the coil B_x

Experimental B_x =(T)

Theoretical B_x =(T)

